LESSON 5.2

Determine the Area of Triangles

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the area of a triangle using the formula A = ½ × base × height.

WHY THIS MATTERS Your architecture firm is designing a park with triangular garden beds. The client wants to know exactly how much soil to order, which means calculating the area of each triangular section. The first garden has a base of 12 feet and a height of 8 feet.

REMEMBER

Determine the Area of Triangles

Area of Triangles. Formula: Area = ½ × Base × Height (A = ½ b · h) A triangle has three sides. The base is one side, and the height is the straight-up (perpendicular) distance from the base to the top corner.

1. Identify the base and the perpendicular height to that base.
2. A triangle is exactly half of a rectangle with the same base and height.
3. Multiply base by height, then divide by 2

Words: **Base *(Base)*** · **Height *(Altura)*** · **Area *(Área)*** · **Perpendicular *(Perpendicular)*** · **Composite figure *(Figura compuesta)***

1 **Circle the letter of the best answer. Show how you know.**

What is the area of a rectangle 8 cm long and 5 cm wide?

(A) 13 square cm
(B) 26 square cm
(C) 40 square cm
(D) 40 cubic cm

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

2 **Circle the letter of the best answer. Show how you know.**

What is the area of a parallelogram with base 9 in and height 4 in?

(A) 13 square in
(B) 18 square in
(C) 26 square in
(D) 36 square in

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

3 **Circle the letter of the best answer. Show how you know.**

In a parallelogram, how is the height measured?

(A) Along the slanted side
(B) Perpendicular to the base
(C) Around the outside
(D) From one corner to the opposite corner

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A parallelogram has an area of 48 square cm and a base of 8 cm. What is its height?

(A) 6 cm
(B) 40 cm
(C) 56 cm
(D) 384 cm

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

5 **Circle the letter of the best answer. Show how you know.**

A homeowner wants to paint one triangular wall of an A-frame cabin. The wall has a base of 20 feet and a height of 14 feet. One can of paint covers 50 square feet. What is the area of the triangular wall?

(A) 34 sq ft
(B) 70 sq ft
(C) 140 sq ft
(D) 280 sq ft

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

6 **Circle the letter of the best answer. Show how you know.**

A homeowner wants to paint one triangular wall of an A-frame cabin. The wall has a base of 20 feet and a height of 14 feet. One can of paint covers 50 square feet. How many cans of paint does she need?

(A) 2
(B) 2.8
(C) 3
(D) 4

SHOW YOUR WORK

EXPLAIN YOUR THINKING

The triangle fits inside a 12 × 8 rectangle. How does the triangle's area compare to the rectangle's area?

SENTENCE STARTER The area of the triangle is ___ the rectangle because A = 1/2 × ___ × ___ = ___ square feet.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 Write the letter of the matching item on each line.

Match each triangular plan on the blueprint to its area. Use A = ½ × base × height.

_____ Triangle base 10, height 6	(A) 30 sq ft
_____ Triangle base 7, height 4	(B) 14 sq ft
_____ Triangle base 9, height 8	(C) 36 sq ft
_____ Triangle base 16, height 3	(D) 24 sq ft
_____ Triangle base 5, height 8	(E) 20 sq ft
_____ Triangle base 11, height 2	(F) 11 sq ft

2 Write the name of the correct group on each line.

An intern at the firm gave two answers for each triangular wall. Sort each answer into the CORRECT area (½ × base × height) or the FORGOT THE HALF mistake (just base × height).

Groups: **Correct area (½ × base × height)** **Forgot the ½ (just base × height)**

_____ Wall base 18, height 6 → 54 sq ft

_____ Wall base 18, height 6 → 108 sq ft

_____ Wall base 20, height 5 → 50 sq ft

_____ Wall base 20, height 5 → 100 sq ft

_____ Wall base 14, height 8 → 56 sq ft

_____ Wall base 14, height 8 → 112 sq ft

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A homeowner wants to paint one triangular wall of an A-frame cabin. The wall has a base of 20 feet and a height of 14 feet. One can of paint covers 50 square feet. Why round up instead of down?

(A) Because 2 cans cover only 100 sq ft, which is not enough for 140

(B) Because paint is always sold in threes

(C) Because 2.8 rounds to 3 by the usual rule

(D) Because the wall might be bigger

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A triangular window in the firm's lobby has a base of 10 ft and a height of 6 ft. What is its area?

(A) 16 sq ft

(B) 30 sq ft

(C) 32 sq ft

(D) 60 sq ft

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

5 **Circle the letter of the best answer. Show how you know.**

A triangular roof panel on the blueprint has a base of 7 ft and a height of 4 ft. What is its area?

(A) 11 sq ft

(B) 14 sq ft

(C) 22 sq ft

(D) 28 sq ft

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

6 **Circle the letter of the best answer. Show how you know.**

On the blueprint, a triangular garden bed has a base of 11 ft and a height of 6 ft. What is its area?

(A) 17 sq ft
(B) 30 sq ft
(C) 33 sq ft
(D) 66 sq ft

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

EXPLAIN YOUR THINKING

The triangle fits inside a 12 × 8 rectangle. How does the triangle's area compare to the rectangle's area?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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